Gilbert TUYAMBAZE

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Summary

Self-taught software developer and Mathematics, Physics, and Computer Science (MPC) graduate focused on building scalable software systems and technology solutions for underserved environments. Developed full-stack platforms, real-time communication systems, indoor navigation prototypes, automation tools, and educational infrastructure projects using independent research and experimentation. Creator of UR-NAV, a real-time indoor positioning and navigation prototype using Wi-Fi fingerprinting, IMU sensor fusion, Kalman filtering, and graph-based routing. Founder of Twigane Community, a peer-learning initiative expanding access to programming and digital literacy education in Rwanda. Interested in software engineering, distributed systems, intelligent infrastructure, and applied computer science research.

Experience

Secretary, NiCE Club Rwanda (Nuclear is Clean Energy) 2026 – Present

- Support the planning, coordination, and documentation of club meetings, activities, and organizational decisions.
- Participate in nuclear energy advocacy and public outreach initiatives to raise awareness and improve understanding of nuclear science, technology, and its peaceful applications.
- Contribute to educational outreach activities in secondary schools, engaging students and encouraging interest in nuclear science, clean energy, and STEM careers.
- Help organize and participate in events, conferences, and stakeholder engagements promoting nuclear energy awareness, including activities associated with the Nuclear Energy Innovation Summit for Africa (NEISA).
- Facilitate communication among club members, coordinate announcements, and support collaboration on initiatives advancing nuclear education and youth engagement in Rwanda.

Founder and Programming Mentor, Twigane Community Apr 2023 – Present

- Founded and led a peer-learning programming community focused on expanding access to technology education and collaborative learning opportunities.
- Designed mentorship structures including rotating teach-backs, peer debugging sessions, and offline-first learning systems to support students with limited internet access.
- Built and maintained the Twigane online learning platform containing downloadable tutorials, coding exercises, and beginner-friendly educational resources.
- Mentored students in web development, programming fundamentals, and problem-solving through practical project-based learning.

Student Leader – Entertainment and Electronics Department, ES Bumbogo Apr 2022 – Mar

- Coordinated school entertainment and electronics activities including audio systems and event organization.
- Assisted in managing technical equipment used during student events, announcements, and school activities.
- Collaborated with student leaders and staff to improve communication and event coordination.

Anti-Corruption Representative, ES Bumbogo Jan 2022 – Mar 2023

– Represented students in promoting accountability, transparency, and responsible conduct within school activities.

– Participated in awareness initiatives and discussions focused on ethical leadership and community responsibility.

Projects

URNAV – University of Rwanda Indoor Navigation System (Prototype) Live Demo

Developed a real-time indoor positioning and navigation prototype for the University of Rwanda using Wi-Fi fingerprinting, IMU sensor fusion, Kalman filtering, and A* pathfinding algorithms. Engineered a multi-layer React Native system capable of estimating user position in (x, y, floor) coordinates and generating multi-floor navigation routes without requiring additional hardware infrastructure. Implemented offline-capable positioning workflows, graph-based routing systems, real-time blue-dot tracking, and scalable indoor mapping architecture optimized for low-resource deployment environments.

GRACE-1 – Space-Based Solar Power (SBSP) *2026–Present*

Participate in GRACE-1, a multidisciplinary engineering research initiative under the IEEE University of Rwanda Student Branch, as part of the MTT-S / AP-S technical team focused on antennas and electromagnetic systems.

– Study the system architecture for space-based solar power, tracing the energy-conversion chain from solar collection and DC/RF conversion through microwave transmission and ground-side RF-to-DC conversion.

– Analyze antenna systems, electromagnetic propagation, wireless power transfer, and receiving rectenna concepts relevant to the proposed power link.

– Develop preliminary power-budget and end-to-end efficiency calculations to examine conversion losses and estimate the input power required to deliver a target electrical output.

– Investigate engineering trade-offs involving transmission efficiency, propagation losses, receiving-system performance, and utility-scale power delivery.

MarketplacePro – Multi-Vendor Marketplace Platform Live Demo

Engineered a full-stack multi-vendor marketplace platform supporting administrators, sellers, editors, and customers with role-based access control. Implemented messaging systems, claims management workflows, analytics dashboards, notifications, and order tracking features while developing scalable application architecture, authentication systems, and complex multi-user workflows.

Ournests – Cross-Platform Real-Time Messaging System Live Demo

Built a real-time messaging platform with web, desktop, and mobile clients using Socket.IO, JWT authentication, and MongoDB. Implemented secure authentication, file sharing, live communication workflows, and cross-device synchronization while designing scalable messaging architecture and user privacy systems.

UR Academic Resource Hub Live Demo

Developed an academic resource-sharing platform for university students featuring searchable study materials, upload systems, moderation workflows, and backend APIs built with FastAPI and SQLAlchemy. Designed scalable educational content management systems optimized for accessibility and efficient resource discovery.

Nyagatare Secondary School Website (React, TypeScript) Live Demo

Designed and deployed a responsive school website improving access to school information, announcements, academic programs, and events. Built using React, TypeScript, and Vercel with optimization for low-bandwidth environments and older mobile devices while maintaining lightweight deployment architecture.

Twigane Community Learning Platform Live Demo

Built and maintained an educational platform providing coding tutorials, downloadable learning materials, and offline-accessible programming resources for beginner students. Designed the platform with minimal resource requirements to support accessibility in low-connectivity environments.

Automated School Bell & Announcement System (Visual Basic) 2023

Developed a Visual Basic-based automated school bell and voice announcement system using scheduled audio playback and text-to-speech functionality. Designed the system to replace manual bell operations with human-like voice announcements for schedules, reminders, and school communications while improving operational efficiency.

OrangeMedia – Audio-First Farmers Communication Platform Live Demo

Designed a multilingual communication platform enabling farmers to share audio announcements, low-bandwidth listings, and market information through mobile-friendly interfaces. Built lightweight communication workflows optimized for underserved and low-connectivity environments.

Orangeflixx – Media Streaming Platform Prototype Live Demo

Developed a streaming platform optimized for low-end devices and constrained bandwidth environments. Implemented adaptive interfaces, metadata organization systems, responsive browsing workflows, and lightweight media presentation techniques for accessible user experiences.

Technical Exploration

- Practiced algorithmic problem-solving and computational thinking using platforms including LeetCode, Codeforces, HackerRank, and GeeksforGeeks.
- Strengthened understanding of data structures, recursion, graph traversal, optimization techniques, time complexity, and space complexity through independent problem-solving.
- Explored full-stack architecture, real-time communication systems, secure authentication workflows, and scalable application design through self-directed software development projects.
- Explored indoor positioning systems, sensor fusion, graph-based navigation, and probabilistic filtering through development of the URNAV indoor navigation prototype.

Education

2025 - Present	**University of Rwanda** . CST . BSc(Hons) in Applied Physics	(Year 2)
2024	Rwanda National Advanced Secondary Examination at **NSS**	(70.9%)
2023 – 2024	Class 12th Nyagatare Secondary School (NSS)	Annual Average (71.2%)
2022 – 2023	Class 11th Nyagatare Secondary School (NSS)	Annual Average (81.8%)
2021 – 2022	Class 10th E.S Bumbogo	Annual Average (79.8%)
2020 – 2021	Class 9th G.S Cyabayaga	O'Level (16/72 Aggregate) (Division I)

Achievements

- Selected to serve as Secretary of NICE Club, supporting organizational leadership, meeting documentation, and member communication.
- Founded Twigane Community, a peer-learning initiative helping students access programming education and collaborative technology learning.
- Built and deployed multiple full-stack web applications independently under limited-resource conditions using self-taught development skills.

- Practiced competitive programming and algorithmic problem-solving across platforms including LeetCode, Codeforces, HackerRank, and GeeksforGeeks.
- Developed practical systems including automation software, indoor navigation systems, encrypted communication platforms, educational tools, and marketplace architectures.
- Designed technology solutions optimized for low-bandwidth and low-device environments common in underserved communities.

Skills

Programming Languages	JavaScript, TypeScript, Python, HTML, CSS, Visual Basic
Frameworks & Tools	React, React Native, Node.js, Git, GitHub, Vercel, Supabase, FastAPI, SQLAlchemy
Technical Areas	Full-Stack Development, Real-Time Systems, Indoor Positioning Systems, Algorithmic Problem Solving, System Design, Authentication Systems, Scalable Application Architecture
Algorithms & Concepts	Data Structures, Graph Traversal, A* Pathfinding, Sensor Fusion, Kalman Filtering, Time & Space Complexity Analysis
Soft Skills	Leadership, Mentorship, Community Building, Communication, Independent Learning
Research Areas	Space-Based Solar Power, Wireless Power Transfer, Electromagnetic Propagation, Antenna Systems, Microwave/RF Engineering
Scientific Analysis	Power Budgeting, Energy Conversion Efficiency, System-Level Energy Analysis